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Technical University of Applied Sciences Würzburg-Schweinfurt
Faculty of Computer Science and Business Information Systems

Master's Thesis

Explainable Causer Attribution for Quality Incidents at BMW Group: Knowledge Graph Link Prediction with Evidence-Based Explanations for Automotive Production

Submitted to the Technical University of Applied Sciences Würzburg-Schweinfurt, Faculty of Computer Science and Business Information Systems, in partial fulfilment of the requirements for the degree of Master of Science (M.Sc.) in Artificial Intelligence

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Abstract

When a defect is identified in automotive production, the quality front desk must decide within minutes who caused it: a supplier or an internal unit. This first decision starts the entire follow-up process, including the supplier complaint and the 8D problem-solving process. At BMW Group, the decision is supported by an exact-match search over the defect type, defect place, defect location, part number and supplier, and it is confirmed only weeks later, when the supplier acknowledges or rejects responsibility in the 8D. When no identical historical case exists, the search returns nothing, near-miss cases contribute no evidence, and wrong first decisions lead to misdirected 8D processes, supplier disputes and avoidable cost. This thesis formulates first-call causer attribution as semi-inductive link prediction on an event-centric knowledge graph built from BMW's quality notification, inspection and rework data: each new incident is an unseen node connected to known defect, part, supplier and rework nodes, and the task is to predict its missing `CAUSED_BY` link. Three model families are compared: a rule-based baseline that reproduces the current exact-match search, knowledge graph embeddings (TransE, ComplEx) and a relational graph convolutional network (R-GCN). Test incidents are grouped into novelty levels, from incidents with an exact historical match to incidents whose attribute combination has never been observed. For each family, the thesis measures both predictive performance and the faithfulness of its explanations, using a shared protocol that tests whether the cited evidence is necessary and sufficient for the prediction. The findings are validated on a synthetic event-centric graph with planted causer rules, where the true explanation of every prediction is known. TODO: key quantitative results per novelty level (accuracy and fidelity) and the main conclusion for early causer attribution at BMW.

Zusammenfassung

Wird in der Fahrzeugproduktion ein Fehler festgestellt, muss der Qualitäts-Frontdesk innerhalb weniger Minuten entscheiden, wer ihn verursacht hat: ein Lieferant oder eine interne Einheit. Diese erste Entscheidung stößt den gesamten Folgeprozess an, einschließlich der Lieferantenreklamation und des 8D-Prozesses. Bei der BMW Group wird sie durch einen exakten Abgleich von Fehlerart, Fehlerort, Fehlerstelle, Teilenummer und Lieferant unterstützt und erst Wochen später bestätigt, wenn der Lieferant im 8D-Prozess die Verantwortung anerkennt oder ablehnt. Existiert kein identischer historischer Fall, liefert die Suche kein Ergebnis, ähnliche Fälle bleiben ungenutzt, und falsche Erstentscheidungen führen zu fehlgeleiteten 8D-Prozessen, Lieferantenstreitigkeiten und vermeidbaren Kosten. Diese Arbeit formuliert die Verursacherzuordnung zum Zeitpunkt der Erstentscheidung als semi-induktive Link Prediction auf einem ereigniszentrierten Knowledge Graph, der aus Qualitätsmeldungs-, Prüf- und Nacharbeitsdaten der BMW Group aufgebaut ist: Jeder neue Vorfall ist ein unbekannter Knoten, der mit bekannten Fehler-, Teile-, Lieferanten- und Nacharbeitsknoten verbunden ist; vorhergesagt wird die fehlende `CAUSED_BY`-Kante. Verglichen werden drei Modellfamilien: eine regelbasierte Baseline, die die heutige exakte Suche nachbildet, Knowledge-Graph-Embeddings (TransE, ComplEx) sowie ein relationales Graph Convolutional Network (R-GCN). Die Testvorfälle werden nach Neuheitsgrad gruppiert, von Vorfällen mit exaktem historischem Gegenstück bis zu Vorfällen mit bisher nie beobachteter Merkmalskombination. Für jede Modellfamilie werden die Vorhersagegüte und die Treue (Faithfulness) der Erklärungen gemessen, mit einem einheitlichen Protokoll, das prüft, ob die angeführte Evidenz für die Vorhersage notwendig und hinreichend ist. Die Ergebnisse werden auf einem synthetischen ereigniszentrierten Graphen mit eingebetteten Verursacherregeln überprüft, bei dem die wahre Erklärung jeder Vorhersage bekannt ist. TODO: zentrale Ergebnisse je Neuheitsgrad ergänzen.

Acknowledgements

TODO

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1. Introduction

1.1. Motivation

1.1.1. Industry Motivation: BMW Quality Management

When a defect is identified at an inspection point in production, the first question is who caused it. At BMW Group, this question is answered within minutes of detection by the Quality Management Technology (QMT) front desk. The answer is usually given verbally first and then recorded as the initial causer allocation. It decides everything that follows: a supplier-related defect leads to a quality notification in Qcube (SAP), a supplier complaint and the 8D problem-solving process, whereas an internal defect is passed to assembly, logistics, parts quality or another internal unit for investigation.

This first decision is confirmed only much later. For supplier cases, the confirmation is the supplier's acknowledgement of responsibility, which is recorded in the 8D process, typically during the problem-description stage (D2). Until then, the organization works on the basis of the first decision. If the supplier rejects responsibility, the 8D process was misdirected: time has been spent on supplier communication, sorting and containment, a dispute may follow, and the internal investigation starts late. Conversely, if a genuine supplier defect is first classified as internal, supplier containment is delayed while defective parts may continue to arrive. Both errors increase lead time, rework and cost.

Today, the first decision relies on expert judgment and a structured repeat-failure search. This search looks up historical incidents with the same combination of supplier number, part number and standardized defect description: defect type, defect place and defect location. It is transparent and works well when an identical case has occurred before. However, it returns nothing when no identical case exists, it gives no weight to cases that differ in only one attribute, and it does not systematically use the rework history. Yet the rework history carries useful evidence: the type of rework activity performed on a defective part is informative about its likely causer, and repeated rework on the same part, or on the same part with the same defect, reveals whether a new incident is a repeat or part of a series.

The aim of this thesis is to support the first decision with a prediction of the causer that the later confirmation will establish, made only from the information available at that moment, and to show the evidence behind it: similar historical incidents, whether the new incident appears to be a repeat or part of a series, and the containment actions and root causes recorded for those historical cases. A reliable early prediction would allow internal defects to be handled internally from the start, allow supplier defects to be booked to the supplier before a long investigation, and flag likely rejections before an 8D process is opened. The objective is to support the front desk, not to replace its judgment.

1.1.2. Scientific Motivation

The situation described above is not specific to automotive quality management. Many organizations receive a stream of new events, each described by a small number of categorical attributes that link it to entities shared with past events, and must decide early who is responsible, while the correct answer is confirmed only later. Examples include the routing of IT incident tickets to resolver teams, the attribution of warranty claims to suppliers, and the assignment of maintenance and incident reports to departments.

Represented as a knowledge graph, each event becomes a node connected to its attribute entities, and predicting the responsible party becomes the prediction of a missing link for a node that did not exist when the model was trained. This is a semi-inductive link prediction task: the event itself is new, but the entities it connects to are known [1]. Three broad families of models can address it. Symbolic methods apply explicit rules, such as the current exact-match search. Tensor-factorization models, or knowledge graph embeddings, learn vector representations in which similar cases lie close together [2, 3]. Message-passing models, such as relational graph convolutional networks, compute a representation of a node from its neighborhood in the graph [4]. Earlier work has shown that rule-based and embedding-based link prediction succeed on different kinds of queries [5].

Two questions remain open for this setting. The first concerns novelty. Standard evaluations report performance averaged over all test queries. In an event-centric graph, however, the novelty of a test event can be graded by how much of its attribute combination has been observed before, which makes it possible to ask how each model family behaves as events move away from known precedents. The second concerns faithfulness. Each model family explains its predictions differently: a rule names the matching historical cases, an embedding model can be explained by the training facts that support its prediction, and a graph neural network by the subgraph that influenced it. These explanations are different objects and are rarely compared under the same measure. Yet an explanation is only useful if it reflects the actual basis of the prediction [6]. A plausible expectation is that explanations agree and remain faithful when an exact precedent

exists, but that more flexible models continue to predict on novel events while their explanations no longer reflect what drove the prediction. In a decision setting in which a person must check the evidence before acting, this matters as much as accuracy.

This thesis therefore studies predictive performance and explanation faithfulness jointly, across levels of novelty, for all three model families. BMW's quality data provide the real-world case. A synthetic event-centric graph with planted causer rules complements it, because only in a controlled setting is the true explanation of each prediction known.

1.2. Problem Statement and Research Questions

The preceding motivation established why reliable causer attribution matters, both for BMW's quality management process and, more generally, for industrial decision-making under uncertainty. Building on this motivation, this section formalizes the resulting problem and derives the research questions that guide the remainder of this thesis.

Causer attribution at BMW is decided at the first call, from a sparse defect description, and confirmed only later through the supplier's acknowledgement in the 8D process or through internal assignment. The current exact-match search supports this decision only when an identical historical case exists, and it ignores partially matching cases and rework patterns. Learned models can use such partial evidence, but their usefulness depends on whether the evidence they present actually reflects the basis of their predictions.

This thesis formulates first-call causer attribution as semi-inductive link prediction on an event-centric knowledge graph: given a new incident node connected only to the information available at the time of the first decision, predict its missing `CAUSED_BY` link to a candidate causer. In this thesis, `CAUSED_BY` denotes the organizational responsibility recorded after confirmation, not a physical cause-and-effect mechanism. The following research questions guide the work.

Main research question. *How accurately can the causer of a newly detected quality incident be predicted at the time of the first causer decision by link prediction on an event-centric knowledge graph, and how faithfully do the explanations of symbolic, tensor-factorization and message-passing models reflect the basis of their predictions as incidents become less similar to historical cases?*

SQ1 – Generalization
How does the predictive performance of a rule-based baseline, knowledge graph embeddings and a relational graph convolutional network change across novelty levels, from

incidents with an exact historical match to incidents whose attribute combination has never been observed?

SQ2 – Faithfulness
How faithful are the explanations produced by each model family, measured with a shared protocol that tests whether the cited evidence is necessary and sufficient for the prediction, and how does their faithfulness change across novelty levels?

SQ3 – Validity
Do the patterns observed on the BMW knowledge graph hold on a synthetic event-centric graph with planted causer rules, in which the true explanation of each prediction is known?

The research questions are tested through three hypotheses:

H1
Message-passing models outperform the rule-based baseline on partially and fully novel incidents, but not on incidents with an exact historical match.

H2
Explanation faithfulness decreases as model expressiveness increases, from rules to embeddings to message passing.

H3
The faithfulness gap between model families widens as the novelty of test incidents increases.

1.3. Contributions of the Thesis

This thesis does not propose a new learning algorithm or a new explanation method. Its contributions lie in the formulation, the evaluation design and the empirical evidence:

1. **Formulation.** First-call causer attribution is formulated as semi-inductive link prediction on an event-centric knowledge graph built from BMW quality notification, inspection and rework data, with an explicit definition of which information is available at the time of the first decision and of how the confirmed causer is derived from the supplier acknowledgement and internal assignment.

2. **Evaluation design.** Test incidents are stratified into novelty levels based on how much of their attribute combination has been observed before, and the explanations of all three model families are converted into sets of graph edges so that their faithfulness can be measured with one shared necessary-and-sufficient fidelity protocol.

3. **Empirical comparison on industrial data.** A rule-based baseline reproducing the deployed repeat-failure search, knowledge graph embeddings and a relational graph convolutional network are compared on real production data with respect to accuracy, faithfulness and their behavior across novelty levels, including how many incidents could be attributed early at a given precision.

4. **Controlled validation.** The same protocol is applied to a synthetic event-centric graph with planted causer rules, which provides ground-truth explanations and tests whether the findings depend on properties specific to the BMW data.

1.4. Thesis Outline

The remainder of this thesis is structured as follows. Chapter 2 introduces BMW's quality process and the role of causer attribution within it, followed by the foundations of knowledge graphs, link prediction, knowledge graph embeddings, graph neural networks and explainable AI. Chapter 3 reviews related work on decision support for quality-incident handling, link prediction on industrial knowledge graphs, the comparison of link prediction model families, semi-inductive and inductive link prediction, and the explanation of link predictions and its evaluation, and derives the research gap. Chapter 4 formalizes first-call causer attribution as semi-inductive link prediction, defines the information available at prediction time, the label derivation and the novelty levels, and describes the data sources and graph schema. Chapter 5 presents the three model families and how each one produces an explanation. Chapter 6 details the experimental setup, including the temporal split, the leakage controls, the novelty-level evaluation, the fidelity protocol and the metrics. Chapter 7 reports and discusses the results, including examples of each model family's explanations, and the limitations of the approach. Chapter 8 answers the research questions and outlines future work.

2. Fundamentals

2.1. BMW's Quality Process and Causer Attribution

BMW's quality management process encompasses the detection and documentation of product defects, the containment of their effects, the attribution of responsibility, and the initiation of appropriate problem-solving activities. In the context of this thesis, the central process-level decision is whether a detected defect is attributable to a supplier or to an internal unit, such as assembly, logistics, or another production-related function. This early attribution determines the subsequent investigation and complaint route. The remainder of this section describes the systems, data, and roles involved in this process, from initial defect detection (Section 2.1.1) to the supplier complaint and 8D process (Section 2.1.8). Figure 2.1 summarizes this process at a high level before each stage is described in detail.

2.1.1. Defect Detection and Recording

The process begins when a defect is identified, typically at an inspection point in production; defects may also surface during rework. A defect may concern, for example, the appearance, function, condition, installation, or completeness of a component. The **International Production System for Quality (IPS-Q)** provides the production-side context for defects and rework records. **Modern Quality Capturing (MQC)** is the mobile IPS-Q application used for defect recording and for the initial causer allocation [7]. This initial allocation is the first causer decision on which this thesis focuses (Section 2.1.4).

A defect record may contain the affected material, supplier, plant, vehicle, detection time, a textual problem description, and standardized defect characteristics. The IPS-Q-related data fields identified for this process include:

- the material number;
- the supplier number;

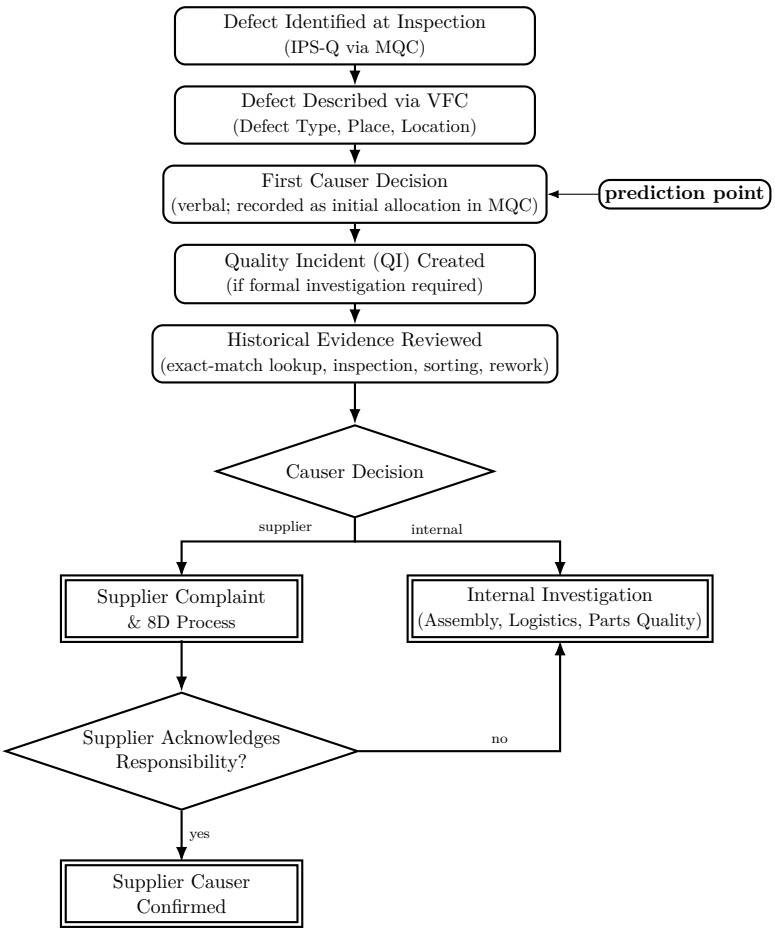


Figure 2.1.: Overview of BMW's quality-incident and causer-attribution process. The first causer decision is made within minutes of detection (prediction point); supplier responsibility is confirmed only later through the supplier's acknowledgement in the 8D process, and a rejection returns the incident to internal investigation.

- the plant;
- the timestamp;
- the defect description; and
- vehicle identification information.

The quality of the initial textual description is important, because it determines whether comparable historical cases can be found and correctly interpreted. The QMT front-desk process explicitly asks whether the problem description is complete and understandable *before* considering repeat occurrence, causer attribution, rework, potential root causes,

and short-term actions. A description such as “door damaged” provides little information for historical comparison, whereas a description such as “damage to the ambient lighting area of the front trim” identifies the affected component and physical position more precisely.

2.1.2. Standardized Fault Coding

To describe defects consistently, BMW uses a standardized fault coding scheme referred to as the **Vehicle Fault Code (VFC; German: *Vereinheitlichte Fehlercodierung*)**. The VFC provides a standardized structure for documenting product-related defects across organizational units, systems, and processes. For this thesis, three VFC elements are particularly relevant: the **defect type**, which describes the observed form of deviation and effect; the **defect place**, which identifies the affected component or functional area; and the **defect location**, which provides a more precise physical or positional localization. Internal data descriptions contain corresponding fields for defect type, defect place, and defect location, which are used together with material, supplier, plant, and vehicle information to identify defect patterns [8].

Table 2.1 illustrates these three elements, together with the shared material key, using the door-trim example introduced above.

Table 2.1.: VFC elements and the shared material key, illustrated using the door-trim damage example.

Element	Description	Example Value
Defect Type	Observed form of deviation and effect	Damage
Defect Place	Affected component or functional area	Door trim, ambient lighting
Defect Location	Precise physical or positional localization	Front right
Material (shared key)	Identifies the affected part	Corresponding door-trim part number

In this thesis, the ordered combination of defect type, defect place, and defect location is referred to as the **defect signature**:

$$D = (T, P, L),$$

where T denotes the defect type, P the defect place, and L the defect location. The defect signature provides a common basis for retrieving related records: the knowledge graph construction described in Chapter 4 likewise treats defect type, defect place, de-

fect location, and material as business keys for linking historical quality incidents and associated rework evidence. In the knowledge graph, the three elements of the defect signature are represented as separate nodes rather than as one combined defect node, so that a new incident that shares only part of its signature with a historical incident remains connected to it. The same decomposition is used in Chapter 4 to grade how novel a new incident is.

A shared defect signature, however, does not by itself establish an identical causer. The damage in the door-trim example could have been present when the component arrived from the supplier, or it could have occurred during internal transport, assembly handling, or rework. The VFC standard describes what was observed; it does not independently determine how the defect arose or who caused it.

2.1.3. Quality Incident Management

When a defect requires formal investigation, it is documented as a **Quality Incident (QI)**. A QI represents the investigation of a quality problem rather than only an individual defect occurrence. Internal process documentation defines a QI as the formal documentation of a defect detected during production, followed by an investigation to identify the responsible unit [9]. The relevant process documentation associates QI creation, messages, and tasks with QCat and Qcube, systems that also support activities for planning and documenting containment, sorting, rework, and inspection.

The distinction between a **defect occurrence** and a QI is important:

- A *defect occurrence* describes a specific observation on a vehicle or component.
- A *QI* provides the broader investigation context in which one or more observations are examined.

For example, the same door-trim damage may be detected on several vehicles; each vehicle may generate a separate inspection or rework record. These records may be investigated under one QI if they are considered manifestations of the same quality problem. Conversely, two defects with the same VFC description may belong to different QIs if they involve different materials, suppliers, plants, or process conditions. The QI therefore serves as the formal context in which a defect, its possible recurrence, and its potentially responsible units are assessed.

Chapter 4 defines whether the unit of prediction in this thesis is the quality incident or the individual defect occurrence.

2.1.4. Process-Level Causer Attribution

Process-level causer attribution refers to the identification of the organizational or process-level unit most likely responsible for a quality defect. The initial decision generally distinguishes between a supplier-related issue and an internally caused issue involving assembly, logistics, or another relevant unit. The QMT front-desk process explicitly includes the question of who the causer for the issue is, with assembly, parts quality, and logistics listed as possible areas to be considered.

Causer attribution must be distinguished from **root-cause identification**. Causer attribution determines the organizational or process-level area to which an incident should initially be directed; root-cause identification determines the precise technical or procedural mechanism that produced the defect. For example, assembly may initially be identified as the likely causer before the technical root cause is found to be an unsuitable installation tool or an incorrect handling sequence. Similarly, a supplier may be identified as the likely responsible party before a manufacturing or packaging deviation is confirmed as the technical root cause.

The QMT front-desk assessment includes more than the defect classification. The documented business questions address:

- whether the problem description is complete;
- whether the issue has occurred previously (repeat occurrence);
- who the likely causer is;
- whether the rework has been fully and correctly recorded;
- which potential root causes and short-term actions are known;
- whether the issue is isolated or part of a series; and
- whether other plants are, or may be, affected.

The result is an early process decision, made within minutes of detection on the basis of the information available at that time. It is usually given verbally first and then recorded as the initial causer allocation. It determines the follow-up route, but it is provisional: for supplier cases, responsibility is confirmed only when the supplier acknowledges it in the 8D process (Section 2.1.8), and a rejection returns the incident to internal investigation. This thesis predicts the causer at the time of this first decision. Root causes and containment actions are not predicted; they appear only as historical evidence attached to similar past incidents.

2.1.5. Historical Lookup and Repeat-Problem Identification

The current investigation process combines structured historical lookup with expert assessment. Internal QMT training material describes a repeat-failure search that queries three different combinations of supplier number, part number, and VFC attributes [9]. An empty result list indicates that no comparable QI was found using the active search configuration; where a QI with the same defect pattern already exists, the new QI can be linked to the original case and marked as a repeat failure. The rule-based baseline in Chapter 5 reproduces exactly these search combinations, so that it represents the deployed process rather than an idealized rule system.

At a high level, the current process can be summarized as follows:

1. A defect is recorded in IPS-Q through MQC.

2. The defect is described using the VFC and relevant product and process information.

3. A QI is created when formal investigation is required.

4. Historical incidents are searched using configured combinations of supplier, part, and VFC attributes.

5. Related inspection, sorting, and rework information is reviewed.

6. The incident is assigned to a supplier or an internal unit.

7. The responsible expert validates the assignment and initiates the corresponding follow-up process.

This sequence is a synthesis of the documented system landscape and front-desk business questions; the exact execution may vary according to the type and context of the incident. For example, an investigator assessing door-trim damage may retrieve incidents involving the same material and VFC description. If comparable incidents were previously assigned to the supplier, they provide historical support for supplier attribution. Nevertheless, their applicability to the current case must be evaluated using the supplier, plant, detection point, production stage, and other available context.

The current lookup is transparent, because the filter attributes are explicit. Its coverage is nevertheless limited by the defined search combinations and the completeness of the recorded information: a relevant historical case may not be returned if it differs in material, supplier, fault type, or another required attribute. This exact-match limitation is the central weakness of the current process that motivates this thesis.

2.1.6. Inspection, Sorting, and Rework

Inspection, sorting, and rework are related quality activities, but they serve different purposes and produce different forms of information. **Inspection** evaluates a product's quality characteristics and records the context in which a defect was identified; the inspection information considered in this thesis includes the station, the production stage, the relevant quality check, and the test result. **Sorting** examines a defined population of parts or vehicles, and it distinguishes sorting performed by a supplier, internally by BMW, or through a mixed form involving BMW personnel, a quality service provider, and the supplier. **Rework** documents how a detected defect was corrected; the relevant records may contain the defect type, the repair action (referred to as the rework activity), the affected component, the plant, and the responsible department.

Two aspects of rework are particularly relevant for causer attribution. First, the type of rework activity carries information about the likely causer, because different corrections are typical for different defect origins. Second, rework occurrences accumulate on parts: repeated rework on the same part, or on the same part with the same defect signature, reveals whether a new incident is a repeat or part of a series.

Across the different systems used by QMT, these activities are represented through **Sorting, Rework, and Inspection (SRI) orders**. The documented SRI process includes order creation, instructions, and the recording of inspection results as OK or NOK (not OK).

These activities provide complementary perspectives: inspection records where and under which conditions a defect is detected, sorting records the result of examining a defined population, and rework records how the defect was corrected. These perspectives must not be confused with causation. The place at which a defect is detected is not necessarily the place at which it was created; likewise, the unit that performs the repair is not automatically the causer. For example, door-trim damage detected at an assembly inspection point may already have existed before the component entered assembly; conversely, a conforming component may be damaged during installation and subsequently reworked at another station. Detection, correction, and causation therefore remain separate aspects of the investigation.

Rework information is therefore treated as evidence, not as proof of causation. Whether the rework record of a new incident is available at the time of the first causer decision depends on when it is recorded; Chapter 4 defines which records may be used for prediction.

2.1.7. Roles in the Quality Process

Several roles contribute to defect detection, investigation, containment, and problem-solving:

Production and Assembly
Identify defects in the manufacturing environment and provide the initial defect and process context. MQC is positioned in the internal ecosystem as the application for recording a defect and performing an initial allocation to a likely causer.

Quality Engineers and the QMT Front Desk
Examine the problem description, compare the issue with previous cases, assess a likely causer, and determine the appropriate follow-up process. Their documented business questions include repeat occurrence, causer identification, rework completeness, potential root causes, short-term actions, and affected plants.

Internal Units (Assembly, Logistics, Parts Quality, and Related Units)
May receive incidents for further investigation when internal responsibility is suspected. These units appear as potential causer categories in the processing paths documented in the QMT front-desk material.

Suppliers
Participate in the complaint and problem-solving process when supplier responsibility is suspected. A dedicated supplier portal, qWin, supports the exchange and management of supplier-related 8D information, including requested 8D reports. External quality service providers may perform sorting activities, while mixed sorting additionally involves the supplier.

2.1.8. Supplier Complaints and the 8D Process

If a defect is considered supplier-related, the incident can proceed to a supplier complaint process. The documented system landscape connects the quality notification to **8D (eight-disciplines) reporting**.

The 8D method is a structured problem-solving approach used to identify, correct, and prevent recurring product or process problems. It progresses from team formation and problem definition, through containment and root-cause analysis, to permanent corrective actions and the prevention of recurrence.

The 8D method originated in the automotive industry as a standardized framework for supplier and internal problem-solving and is applied within BMW's quality management process when supplier responsibility is suspected; confirmation of that responsibility is itself part of the process. Although the disciplines are numbered sequentially, they are not always executed strictly one after another in practice: D3 (interim containment) is typically initiated in parallel with, or even before, the completion of D4 (root-cause analysis), since protecting the customer and the production process cannot wait until the root cause is fully understood.

A central element for this thesis is the supplier acknowledgement, a dedicated field in the 8D process in which the supplier accepts or rejects responsibility for the defect. The acknowledgement typically follows the problem description (D2). Until it is given, the organization operates on the basis of the first causer decision. If the supplier rejects responsibility, the effort already invested in supplier communication, sorting and containment was misdirected, a dispute may follow, and the internal investigation begins late. Predicting the outcome of this acknowledgement at the time of the first decision is the practical goal of this thesis.

In practice, each QI is assigned a priority level, and the eight disciplines are governed by corresponding target timelines that differ between high- and low-priority cases.

The cross-functional team formed under D1 typically draws on the roles introduced in Section 2.1.7: the responsible quality engineer coordinates the investigation, while representatives from production, assembly, and, where relevant, the supplier contribute the domain-specific knowledge needed to describe the problem, contain it, and identify its root cause.

The 8D reports generated through this process also complement the historical lookup described in Section 2.1.5: whereas that lookup retrieves comparable past QIs based on supplier, part, and VFC attributes, an attached 8D report reveals not only that a comparable incident existed, but how it was investigated and resolved, and through which discipline-specific findings.

Each discipline is documented with a responsible owner, supporting evidence, and a completion date. This staged documentation means that an 8D report's evidentiary value depends on how far the investigation had progressed at the time it is reviewed: a report that is still open may contain only a preliminary problem description and containment action, whereas a closed report additionally documents a verified root cause and validated corrective actions.

For the purposes of this thesis, this discipline structure matters because an 8D report is not a single, undifferentiated text block: each discipline corresponds to a distinct, dated piece of evidence, such as a problem description, a containment action, or a verified root cause, that can in principle be linked to the underlying QI as a separate item of evidence

Table 2.2.: The eight disciplines (8D) of the problem-solving process.

Discipline	**Title**	**Purpose**
D1	Establish the Team	Form a cross-functional team with the product and process knowledge required to investigate and resolve the problem.
D2	Define and Describe the Problem	Describe the problem clearly and, where possible, quantitatively. A description should establish what occurred, where and when it occurred, who or what was affected, and the extent of the problem.
D3	Develop and Implement Interim Containment Actions	Introduce temporary measures to isolate the problem and protect the customer or production process while the root cause is being investigated.
D4	Determine and Verify Root Causes	Identify and verify the causes that explain why the problem occurred. This discipline also considers why the defect was not detected earlier. Methods such as the 5 Whys and cause-and-effect diagrams may support the analysis.
D5	Select and Verify Permanent Corrective Actions	Identify permanent actions that address the verified root cause, and confirm that the selected actions are capable of resolving the problem.
D6	Implement and Validate Corrective Actions	Implement the selected permanent corrective actions and verify their effectiveness using appropriate evidence.
D7	Prevent Recurrence	Modify relevant systems, procedures, practices, or controls to prevent the same or similar problems from recurring.
D8	Recognize the Team and Close the Process	Recognize the team's contribution and formally close out the problem-solving process.

rather than treated as one opaque document.

For a new incident, no 8D report exists at the time of the first causer decision; only the 8D reports of historical incidents are available as evidence. Table 2.2 summarizes the eight disciplines reviewed for this thesis.

For causer attribution, the 8D process is particularly relevant because its disciplines produce different forms of historical evidence: D2 contains a structured description of the problem, D3 records the interim containment measures, D4 documents the verified root cause, and D5 through D7 describe the selected, implemented, and preventive corrective actions.

Completed 8D reports can therefore provide evidence about how comparable historical incidents were investigated and resolved. However, a historical 8D report should not automatically determine the causer of a new defect; its relevance depends on whether the current and historical cases are comparable in terms of defect, material, supplier, plant, and process context. For example, an earlier 8D report may document that comparable door-trim damage was related to supplier packaging. This provides relevant historical evidence if the current incident involves the same material, supplier, and incoming inspection point; its relevance is weak if the defect was first identified only after an internal assembly activity. Within this thesis, 8D information of historical incidents is therefore treated as evidence (in particular the recorded containment, D3, and the verified root cause, D4) and not as an automatic attribution rule, while the supplier acknowledgement provides the confirmed causer used as the label (Chapter 4).

2.1.9. Summary

This section has described BMW's current quality process and the data it produces. A defect is first recorded in IPS-Q through MQC and described using the VFC, whose defect type, defect place, and defect location form the defect signature introduced in Section 2.1.2; when formal investigation is required, this is escalated to a QI, which may bundle several defect occurrences or, conversely, split identical-looking defects across several QIs depending on material, supplier, plant, and process context (Section 2.1.3). The central decision at this stage is process-level causer attribution, which must be kept distinct from technical root-cause identification and which the QMT front-desk process supports through a fixed set of documented business questions (Section 2.1.4). Today, this decision relies on a structured but exact-match historical lookup across supplier, part, and VFC attributes, which is transparent but fails whenever no identical historical case exists (Section 2.1.5). Inspection, sorting, and rework records provide complementary evidence about where a defect was detected and how it was corrected, but none of these, nor the unit that performs them, can be equated with causation (Section 2.1.6). Several roles, from production and assembly to quality engineers, internal units, and suppliers, contribute evidence at different stages of this process (Section 2.1.7), and supplier-related incidents additionally generate 8D reports, whose eight disciplines yield further, differently reliable forms of historical evidence depending on how far an investigation has progressed (Section 2.1.8).

Across all eight subsections, the same pattern recurs. The causer is decided early, within minutes and on sparse information, but confirmed late, through the supplier's acknowledgement in the 8D process. The evidence that could support the early decision – historical incidents, rework activities and their repetition, and the outcomes of past 8D processes – is distributed across IPS-Q, QCat, Qcube, SRI orders and 8D reports, and today it is used only through exact-match search on a small set of attributes. This motivates the approach of this thesis: to integrate these sources into one knowledge graph

(Chapter 4) and to predict the causer at the time of the first decision from partially matching historical cases, together with evidence whose faithfulness to the prediction can be measured (Chapters 5 and 6).

2.2. Knowledge Graphs and Link Prediction

A knowledge graph represents knowledge as a network of entities and their relationships. Entities correspond to distinguishable real-world objects or abstract concepts and are represented as nodes, while relations are represented as typed edges connecting these nodes. Unlike conventional data representations that treat records independently, a knowledge graph explicitly captures how different objects relate to one another, providing a structured and semantically meaningful representation of interconnected information [10].

Formally, a knowledge graph can be defined as a collection of facts represented as triples

$$(h, r, t),$$

where h denotes the head entity, r the relation, and t the tail entity [10, 11]. The relation specifies the meaning of the connection between the two entities. For example, the following triples illustrate this structure:

$$\begin{gathered}(\text{Paris}, \text{CAPITAL_OF}, \text{France}) \\ (\text{Component}, \text{SUPPLIED_BY}, \text{Supplier}) \\ (\text{Incident}, \text{HAS_DEFECT_TYPE}, \text{DefectType})\end{gathered}$$

The first states that Paris is the capital of France; the second, that a component originates from a particular supplier; and the third records the type of defect observed in a reported incident.

A knowledge graph is formed by combining many such triples, allowing the same entity to participate in multiple relations. A single quality incident, for instance, may simultaneously be connected to the plant at which it occurred, the defect it reports, the rework performed in response, and the causer ultimately identified for it. The resulting structure is therefore a multi-relational graph, in which different edge types express different semantic relationships [11], as illustrated in Figure 2.2 using the schema proposed for this thesis.

This schema follows an event-centric design: the Quality Incident node represents a discrete occurrence in time, while the remaining entities describe either the circumstances under which it occurred (plant, material, supplier) or the response to it and its eventual resolution (defect, rework, inspection, causer). Each entity type corresponds to a con-

crete record type already introduced in Section 2.1: incidents and defects originate from IPS-Q and the VFC, rework and inspection records from SRI orders, and the `CAUSED_BY` relation from the confirmed causer, that is, the supplier's acknowledgement in the 8D process or the final internal assignment. In this thesis, `CAUSED_BY` denotes recorded organizational responsibility, not a physical cause-and-effect mechanism. Representing them within a single graph keeps the structure extensible, since new incidents, materials, or suppliers can be added as they occur without requiring changes to the relations already defined between existing entity types.

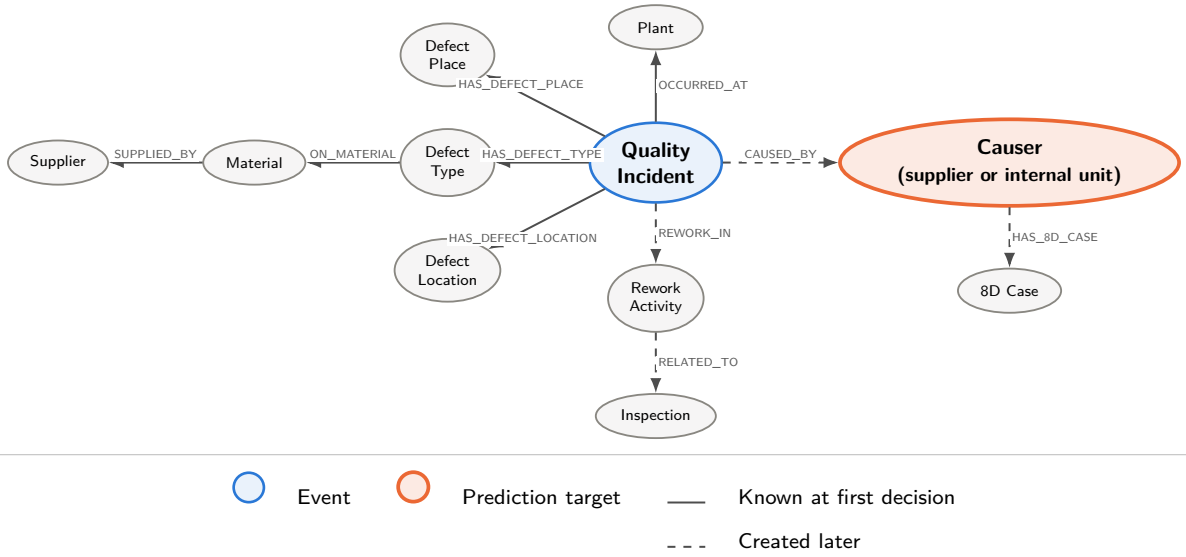


Figure 2.2.: Knowledge-graph schema for causer attribution. The quality incident is the central event node. Solid edges denote information available at the time of the first causer decision; dashed edges denote information created later. The `CAUSED_BY` edge to a candidate causer (a supplier or an internal unit) is the prediction target (highlighted).

Entities and relations may additionally carry types and properties: an incident entity may be described by attributes such as an identifier, a creation date, and a status, while a defect entity may be described by its type, location, and affected area. These descriptive properties characterize individual entities, whereas relations express how entities are connected; together, they allow heterogeneous information to be represented within a common structure while preserving its original domain-specific meaning [10].

This representation is particularly suitable for domains in which relevant information is distributed across multiple sources. In industrial quality management, for example, analyzing a reported incident may require information about defects, materials, plants, suppliers, inspections, rework, and containment actions. These objects describe different aspects of the same underlying context; representing them as connected entities makes

it possible to retrieve relevant information through their relationships, rather than examining each source independently. The same principle applies more broadly: wherever an organization's knowledge is distributed across heterogeneous records that share common objects, a knowledge graph offers a structured way to reconnect that information for retrieval and reasoning.

2.2.1. Knowledge Graph Completion and Link Prediction

Knowledge graphs are generally incomplete, because not every valid relationship is observed or explicitly recorded. The absence of a link does not necessarily imply that no relationship exists; it may instead indicate that the corresponding fact has not yet been identified, entered, or derived from the available information. Knowledge graph completion addresses this incompleteness by identifying plausible missing facts from the existing graph structure [10].

Link prediction is one of the principal tasks used for this purpose. Given an incomplete triple, link prediction estimates the missing entity or evaluates whether a proposed relationship is plausible [11, 12]. The task may take different forms:

$(h, r, ?)$ the tail entity is unknown,
$(?, r, t)$ the head entity is unknown,
(h, r, t) the plausibility of a complete triple is evaluated.

For example, given the incomplete triple

$$(\text{Incident}, \text{CAUSED_BY}, ?),$$

the link prediction task is to identify the most plausible causer among a set of candidate entities. Candidates are typically ranked according to a scoring function that estimates the plausibility of each possible completion; the highest-ranked candidate is the entity considered most compatible with the learned graph patterns. Link prediction therefore transforms the graph completion problem into a structured prediction task over entities and relations [12, 11].

Knowledge graph embedding methods commonly support this task by mapping entities and relations into a continuous vector space. A scoring function then evaluates how plausible a given combination of head entity, relation, and tail entity is; the representations are learned such that observed triples receive higher plausibility scores than incorrect or unobserved combinations. However, the ability of such a model to handle new entities depends on whether it follows a transductive or an inductive learning setting.

2.2.2. Transductive and Inductive Link Prediction

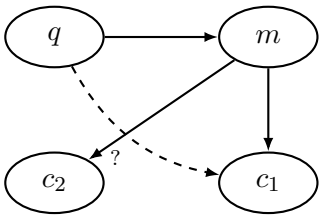
In a **transductive** setting, the set of entities considered during inference is already known at training time: the model observes a graph containing all relevant entities and learns representations for them, even if some of their relationships are hidden and later used for evaluation. Conventional knowledge graph embedding methods frequently follow this paradigm by assigning a trainable embedding vector to each entity in the training graph [11, 13]. Such models can predict missing links between entities for which embeddings have already been learned; for example, if an incident and all of its possible causers were included in the training graph, the model could estimate which of the missing relations between these known entities is most plausible. This setting, however, assumes that the entities required during inference were already available at training time: an entity-specific embedding cannot be retrieved for an entity that did not occur in the training graph [13, 12].

This limitation is significant in dynamic application environments, where new products, incidents, materials, or other business objects may continuously be added after a model has been trained. Retraining the entire model whenever a new entity appears is generally undesirable, and, more importantly, a model that depends only on memorized entity representations does not demonstrate that it has learned relational patterns transferable to previously unseen cases.

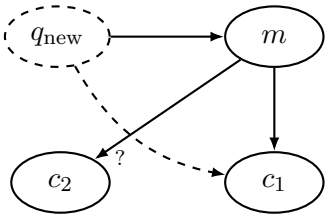
Inductive link prediction addresses these limitations by requiring a model to generalize to entities that were not part of the training graph [13, 14]. During inference, the graph may contain entities absent from training; instead of relying exclusively on a stored embedding for every entity, an inductive method derives information from transferable features, relations, or structural patterns in an entity's graph neighborhood. A neighborhood describes the entities and relations surrounding a target entity; for an incident, this neighborhood may include its associated defect, material, plant, and inspection or rework records. Although the incident itself is new, some of these neighboring entity types and relational structures may resemble patterns observed in historical incidents, and an inductive method can use the structure surrounding the new entity to support its prediction. Subgraph-based inductive approaches formalize this idea by extracting a local subgraph around the candidate entities and learning the relational and structural patterns within such subgraphs, rather than only the identities of individual entities [13]; the learned patterns can then be applied to a new subgraph containing an entity that was absent during training.

Inductive settings differ in how much of the test graph is new. In the *semi-inductive* setting, a new entity appears at inference time but is connected to entities that were present during training; in the *fully inductive* setting, inference takes place on a graph whose entities are entirely disjoint from the training graph [1]. A newly detected quality incident is a new node connected to known defect, part, and supplier nodes, so

causer attribution falls into the semi-inductive setting. Figure 2.3 contrasts the transductive and semi-inductive settings for the causer-attribution task developed further in Section 2.2.3.



(a) Transductive
q, c_1, and c_2 were all present during training.



(b) Semi-inductive
q_{new} (dashed) is absent from training; only its connection to m is available at inference time.

Figure 2.3.: Transductive versus semi-inductive link prediction. In both panels, the dashed edge marks the relation to be predicted. (a) In the transductive setting, the incident q whose causer relation is masked was already present during training, so an entity-specific representation for q has already been learned. (b) In the semi-inductive setting, the new incident q_{new} did not exist during training; the model must instead use its known connection to the material m to judge the plausibility of the candidate triple $(q_{\text{new}}, \text{CAUSED_BY}, c_1)$ relating it to a causer.

The distinction can be summarized as follows: transductive link prediction predicts missing relations among entities that are already known during training, whereas inductive link prediction transfers learned relational patterns to entities that were not present during training. This distinction concerns the availability of *entities*, rather than only the presence or absence of individual edges: hiding a relation for an incident that was already observed during training remains a transductive task, whereas predicting the same type of relation for an incident that was absent from training, but is connected to known entities, constitutes a semi-inductive task.

2.2.3. Relevance to the Proposed Methodology

SQ1 (Section 1.2) asks how well causer attribution generalizes to quality incidents that have no exact historical match. The concepts introduced above give this question a precise formulation: causer attribution is framed as the prediction of a missing relation between a quality incident and a candidate causer,

$$(q, \text{CAUSED_BY}, c),$$

where q denotes a quality incident and c a candidate causer, where the candidate causers are the suppliers and internal units defined in Chapter 4. Historically resolved incidents provide examples for which this relation is already known; a newly created incident has no known causer and must instead serve as the head entity of an incomplete triple,

$$(q_{\text{new}}, \text{CAUSED_BY}, ?),$$

whose missing tail is the causer to be predicted.

Because q_{new} did not exist when the historical training graph was built, no entity-specific representation could have been learned for it – this is precisely the semi-inductive setting described in Section 2.2.2. What remains available is the local graph context of q_{new} at the time of the first causer decision: its connections to its defect type, defect place, defect location, material, supplier and plant, and through them, to historical incidents and their confirmed causers. Rather than depending on the identity of the incident, the methodology developed in this thesis compares this relational neighborhood against the structures of previously resolved cases.

Because a new incident may share all, some or none of its attribute combination with historical incidents, the difficulty of the prediction varies from case to case. Chapter 4 therefore groups test incidents into novelty levels, and Chapter 6 evaluates every model on held-out incidents, ordered in time, separately for each level.

2.3. Knowledge Graph Embeddings and Graph Neural Networks

Section 2.2.1 introduced knowledge graph embedding methods only in general terms. This section briefly introduces the two families of representation-learning methods compared in this thesis: translational and bilinear embedding models, which assign each entity and relation a fixed vector learned during training, and relational graph neural networks, which instead derive an entity's representation from its neighbors in the graph.

The translational family is exemplified by TransE [2], which models a relation as a translation from head to tail,

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t},$$

and is simple and efficient, but struggles with relations that are symmetric or connect one entity to many others [11]. ComplEx [3] addresses this by scoring a triple with complex-valued vectors instead of a single translation, which lets it represent both symmetric and asymmetric relations. Both methods, however, learn a dedicated vector for every

entity seen during training and provide no representation for an entity absent from the training graph, which is why Section 2.2.2 classifies them as transductive. To score a new incident, these models therefore require an adaptation, described in Chapter 5.

Relational graph neural networks instead compute an entity's representation from its neighbors rather than from a fixed lookup table. The relational graph convolutional network (R-GCN) [4] extends the graph convolutional network [15] to multi-relational graphs, combining each entity's representation with information from its neighbors, processed separately per relation type, over several layers. Because this computation draws on an entity's actual neighbors rather than a pre-trained vector, it can in principle be applied to an entity that was never seen during training – provided its initial representation does not depend on its identity (e.g. a vector of known attribute values rather than a learned per-entity embedding).

These three mechanisms – the rule-based exact-match baseline of Section 2.1.5, translational and bilinear embeddings, and relational graph neural networks – form the three model families compared in this thesis (Chapter 5). They trade off generalization and interpretability: the rule-based baseline is fully transparent but generalizes only to an identical historical match; embedding models generalize further through geometric similarity, at the cost of a score that is harder to translate into checkable evidence; and graph neural networks generalize furthest, to incidents never observed before, but their layered message passing makes it least obvious which part of the neighborhood drove a prediction. This trade-off is exactly what the main research question (Section 1.2) asks about, and it motivates evaluating accuracy and faithfulness jointly, across novelty levels, rather than in isolation.

2.4. Explainable Artificial Intelligence

As machine learning models increasingly support consequential decisions, understanding *why* a model produced a given output becomes as important as its accuracy. Explainable artificial intelligence (XAI) studies how to make a model's behavior understandable to a human [16, 17]. The literature commonly organizes explanation methods along two dimensions: **intrinsic** (interpretable by design, e.g. a small rule set) versus **post-hoc** (an explanation produced after the fact for an otherwise fixed model) [16], and **local** (one prediction) versus **global** (the model's behavior overall) [17]. Figure 2.4 summarizes this orientation with representative methods per quadrant.

Producing an explanation is not sufficient on its own: it must also be evaluated. **Faithfulness** describes the degree to which an explanation accurately reflects the reasoning the model actually used, rather than a plausible-sounding story constructed after the fact [6]. For explanations that consist of cited evidence, such as graph edges, faithful-

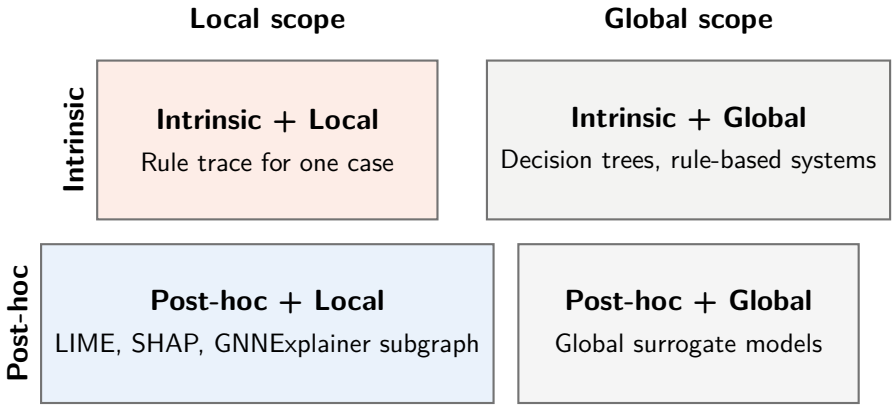


Figure 2.4.: The standard post-hoc/intrinsic and local/global orientation for explanation methods, with representative examples per quadrant [16, 17, 18].

ness can be tested by intervention: an explanation is *necessary* if removing the cited evidence changes the prediction, and *sufficient* if the cited evidence alone still reproduces it. A related property, **sparsity**, concerns how concise an explanation is: fewer, more decisive pieces of evidence are easier to check than an exhaustive list. SHAP [19] is a widely used post-hoc, local method that attributes a prediction to input features via a game-theoretic notion of contribution.

This taxonomy directly informs SQ2 (Section 1.2), which asks how faithfully each of the three model families explains its own predictions. The rule-based baseline is intrinsic and local, since its prediction is the matching rule itself; knowledge graph embeddings and the relational graph neural network are both post-hoc and local, explained respectively through the training facts and the message-passing neighborhood (via GNNExplainer [18]) that most influenced a prediction. Converting all three into a common representation – a set of cited graph edges – is what allows the necessary-and-sufficient intervention test above to be applied uniformly: Chapter 6 defines this shared protocol and applies it to the predictions evaluated in Chapter 7.

3. Related Work

3.1. Root-Cause and Causer Attribution in Manufacturing Quality

TODO

3.2. Research Gap

TODO

4. Problem Formulation and Data

4.1. Formal Problem Definition

TODO

4.2. Data Sources and Graph Schema

TODO

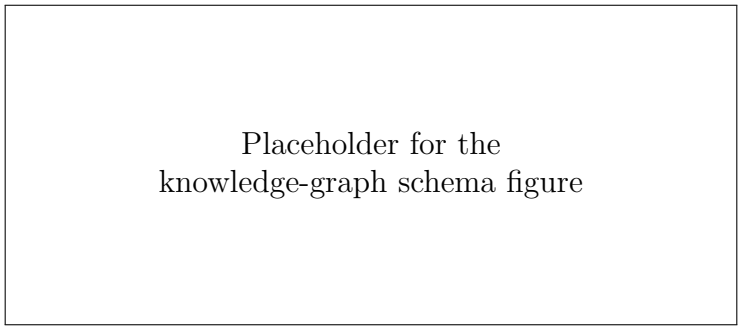


Figure 4.1.: Knowledge-graph schema (placeholder).

As shown in Figure 4.1, TODO.

5. Method

5.1. Model Architecture

TODO

5.2. Explanation Generation

TODO

```
// Retrieve candidate causers for an incident
MATCH (i:Incident {id:
    $incidentId})-[:AFFECTS]->(p:Part)<-[:SUPPLIES]-(s:Supplier)
RETURN s.name AS candidate, count(p) AS evidence
ORDER BY evidence DESC
LIMIT 5
```

Listing 5.1: Example graph query (placeholder).

6. Experimental Setup

6.1. Baselines

TODO

6.2. Evaluation Metrics

TODO

7. Results and Discussion

7.1. Quantitative Results

TODO

Table 7.1.: Comparison with baselines (placeholder values).

Method	Top-1 Acc.	Top-3 Acc.	MRR
Baseline A	–	–	–
Baseline B	–	–	–
Proposed method	–	–	–

7.2. Qualitative Analysis of Explanations

TODO

7.3. Limitations and Threats to Validity

TODO

8. Conclusion and Future Work

8.1. Conclusion

TODO: Answer each research question explicitly.

8.2. Future Work

TODO

A. Additional Tables

Table A.1.: Overview of answer types (placeholder).

Heading 1	**Heading 2**
Title 1	Description 1
Title 2	Description 2

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Declaration of Authorship

I hereby declare that I have written the present Master's Thesis independently and have not submitted it elsewhere for examination purposes. All sources and aids used have been indicated, and all direct and indirect quotations have been marked as such.

Riya Biju, on October 27, 2026

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